Chaos Keepers

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Setup Infrastructure

Platform: Minikube + Docker + Chaos Mesh v2

ZooKeeper Nodes: 3 & 5 nodes

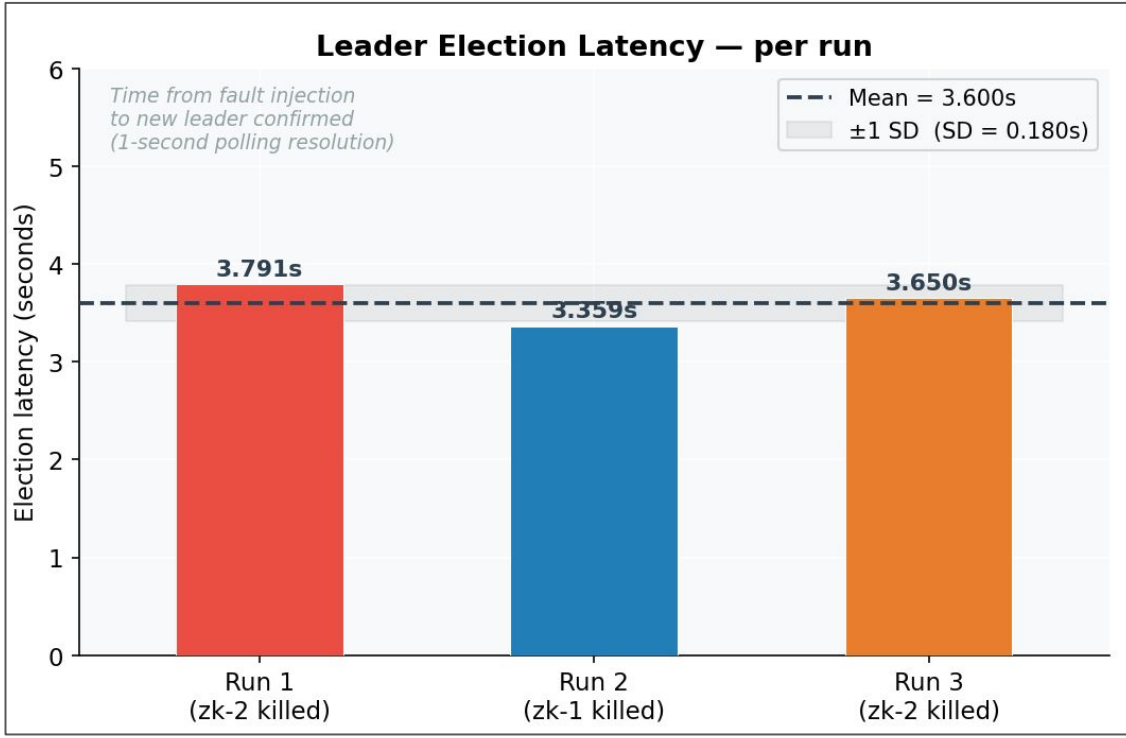
Fault Injection: Chaos Mesh PodChaos - action: pod-kill

- Leader & Followers identified dynamically before each run
- Custom YAML generated and applied (kubectl apply) at injection time
- Fault held for ~60 seconds then removed (kubectl delete)

Monitoring: Python polling every 1 second, queries ZooKeeper endpoints in parallel to detect role changes

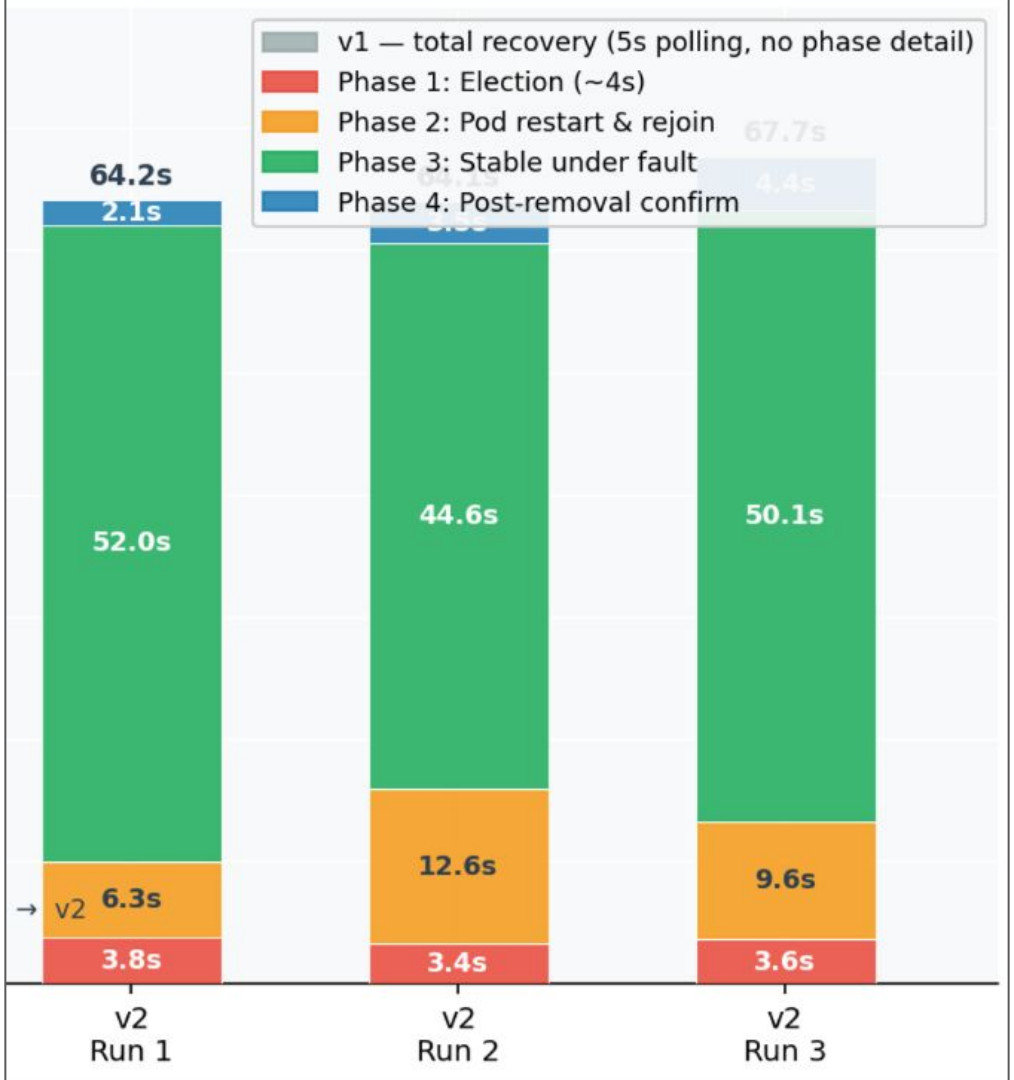
Workload: Concurrent Python client performing write+read ops

Test 1: Kill the Leader

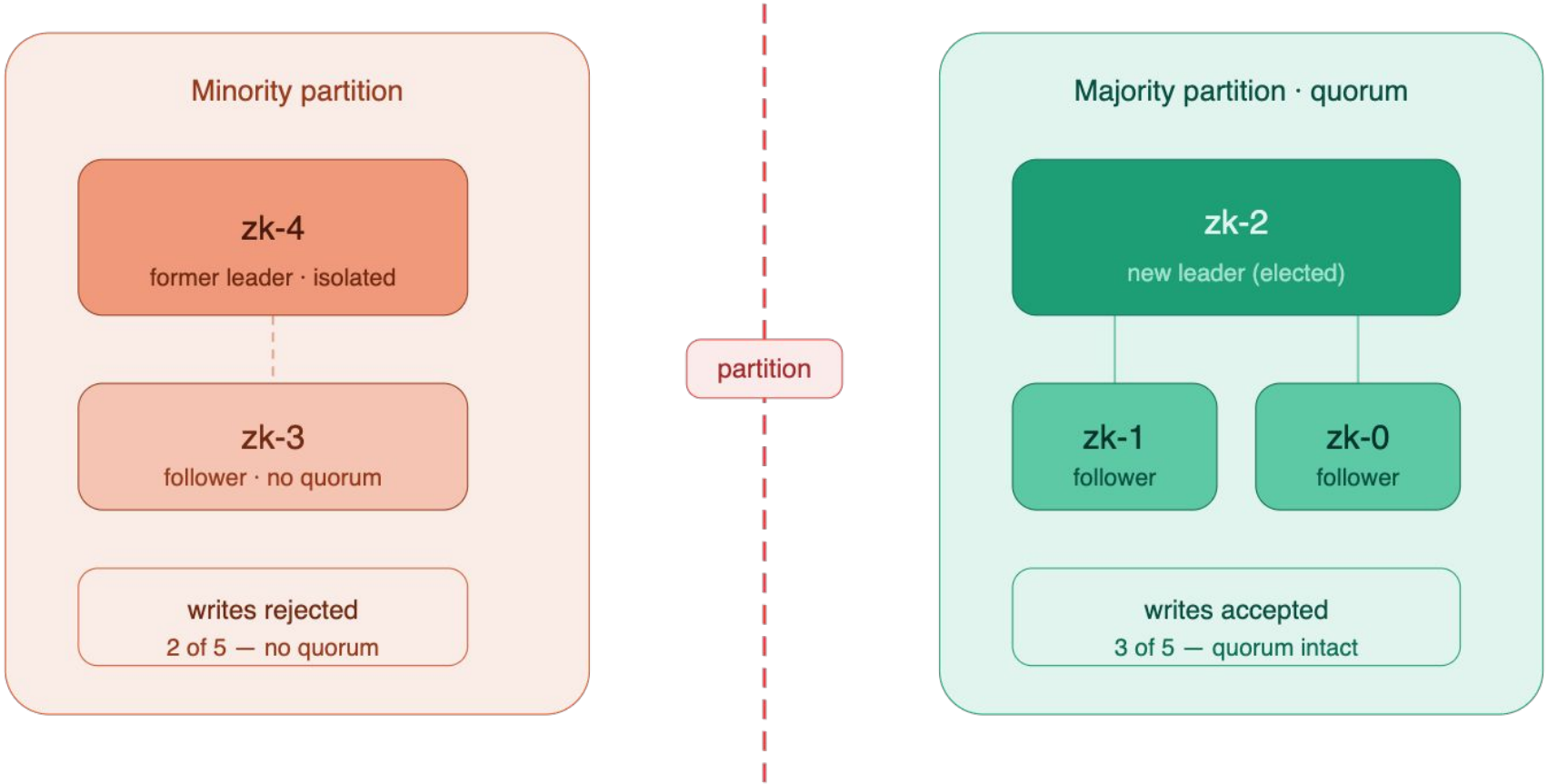


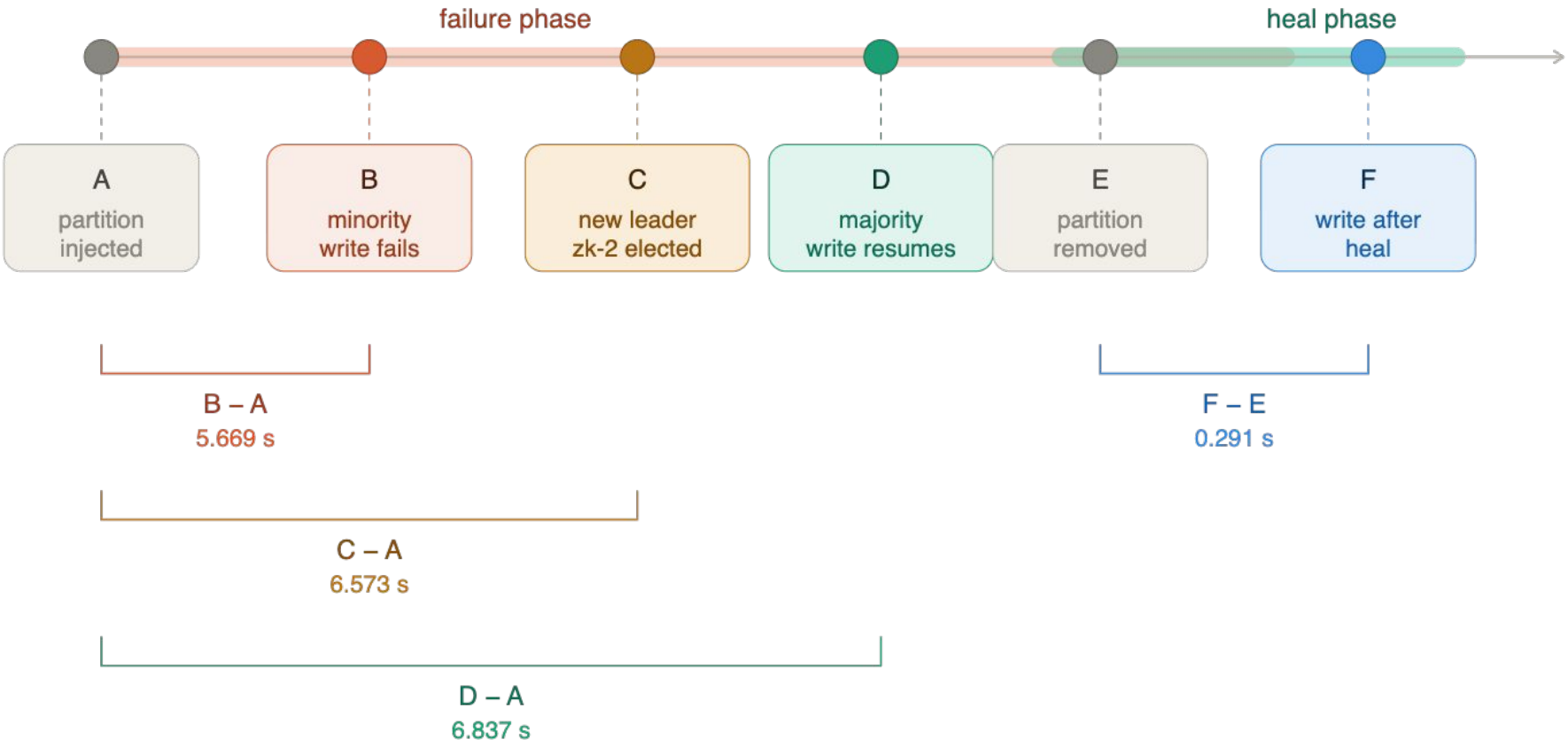
Election Latency Summary

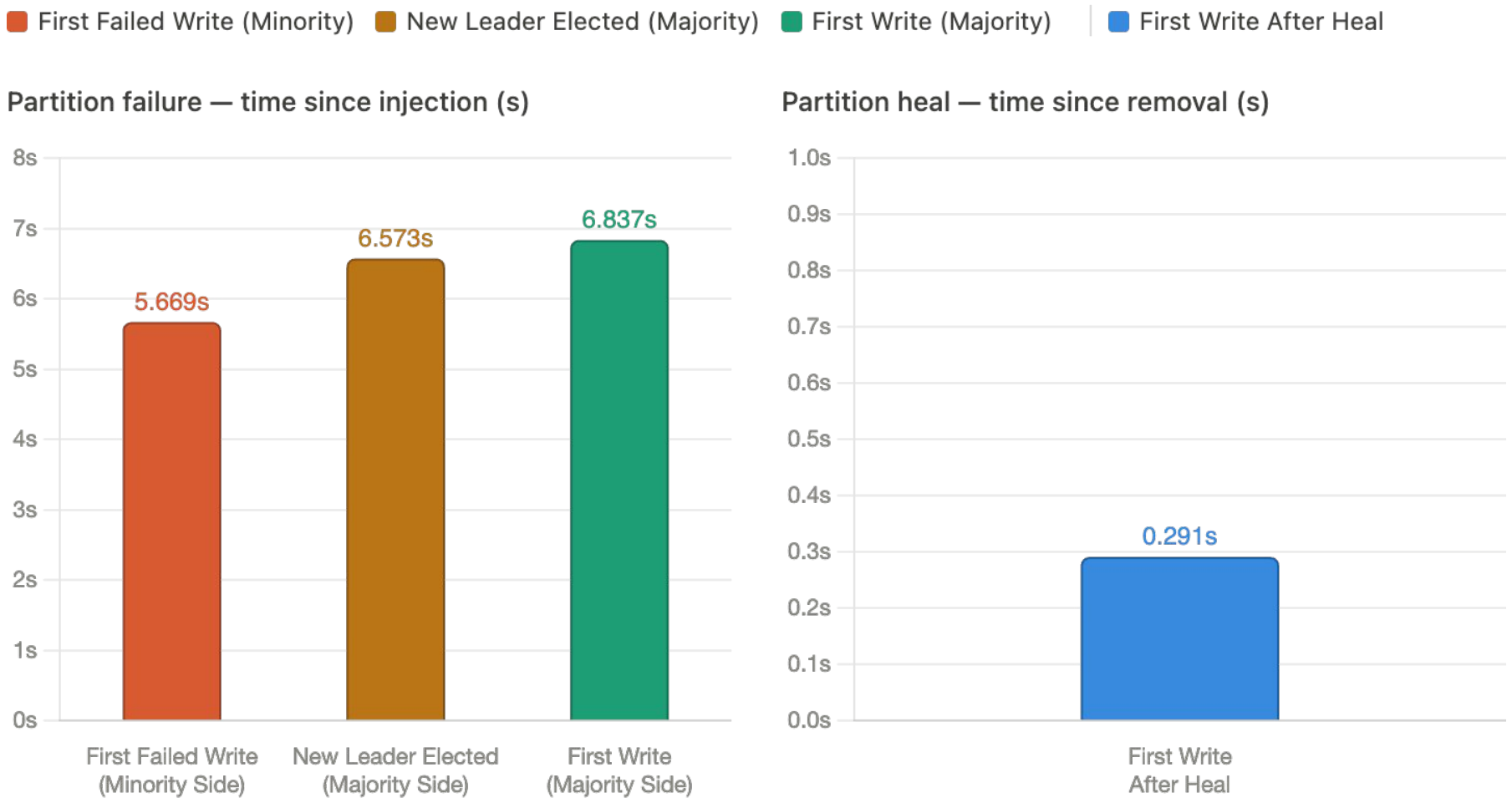
Min:	**3.359 s**
Mean:	**3.600 s**
Max:	**3.791 s**
Std Dev:	**0.180 s**
Runs:	**3 / 3 (100% success)**
Quorum lost?:	**Never (2/3 survive)**


v1 — total recovery (5s polling, no phase detail)
Phase 1: Election (~4s)
Phase 2: Pod restart & rejoin
Phase 3: Stable under fault
Phase 4: Post-removal confirm
64.2s
2.1s
52.0s
6.3s
3.8s
→ v2
44.6s
12.6s
3.4s
50.1s
9.6s
3.6s
v2
Run 1
v2
Run 2
v2
Run 3

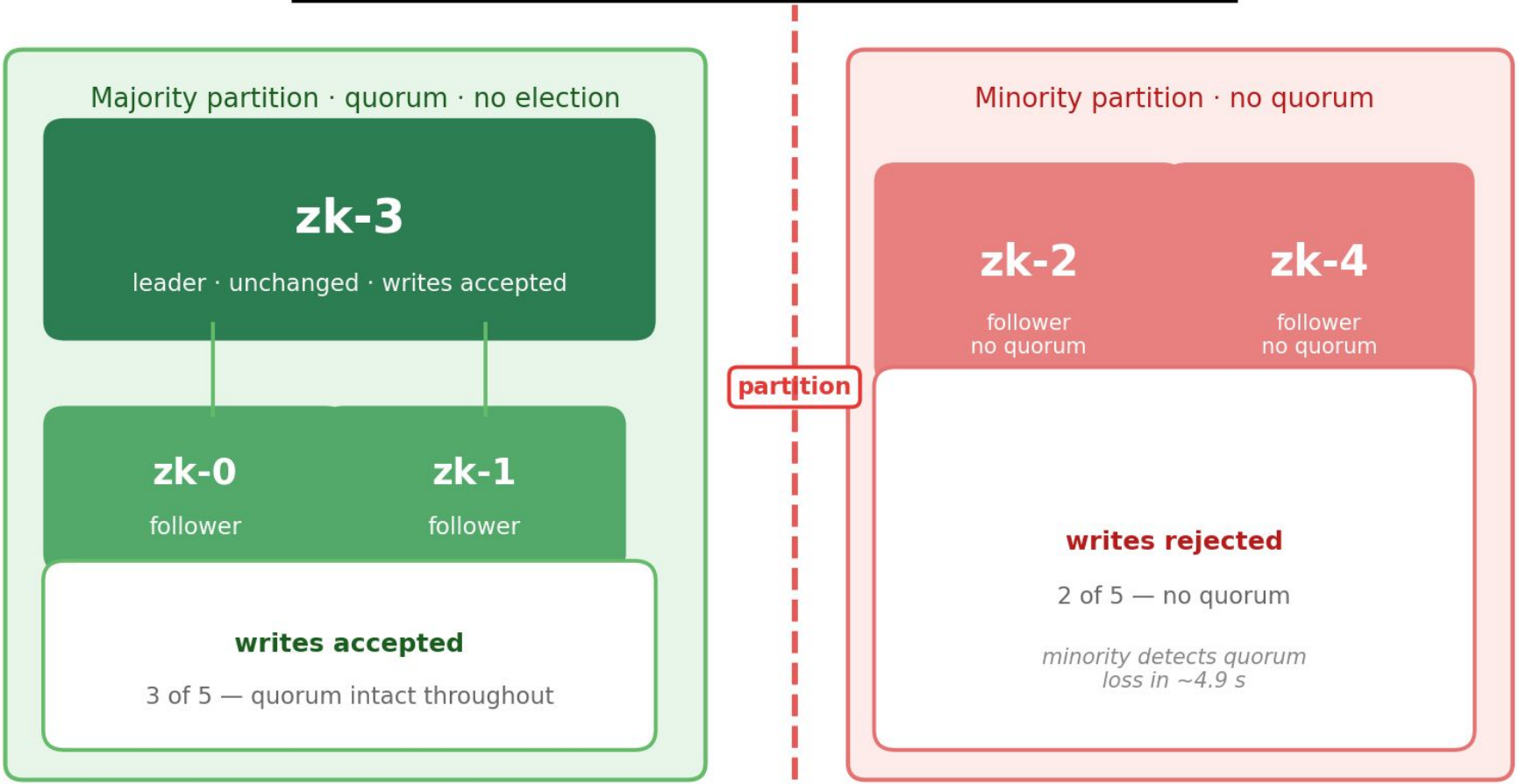
Test 2: System Recovery with Leader on Minority Side

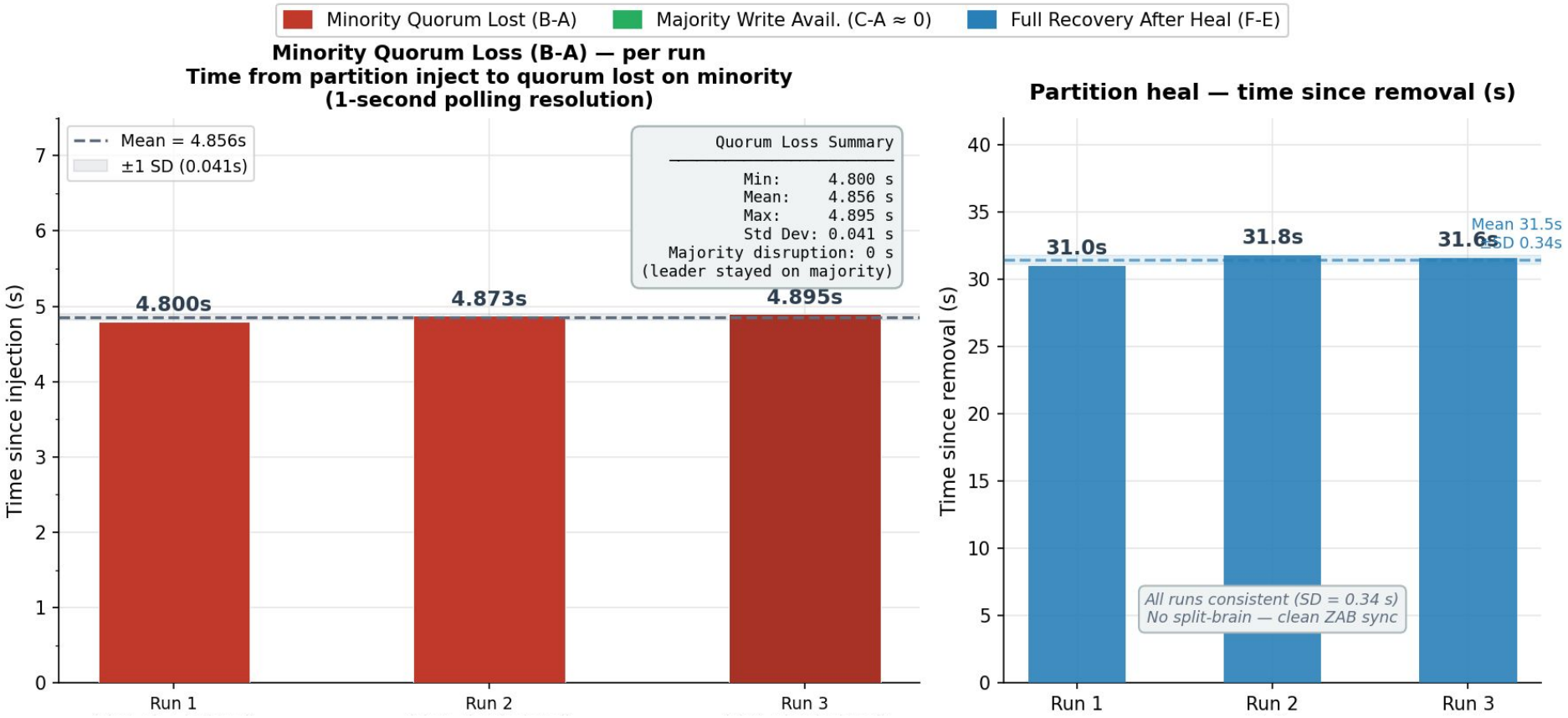



failure phase
heal phase
A
partition injected
B
minority write fails
C
new leader zk-2 elected
D
majority write resumes
E
partition removed
F
write after heal
B – A
5.669 s
F – E
0.291 s
C – A
6.573 s
D – A
6.837 s


First Failed Write (Minority)
New Leader Elected (Majority)
First Write (Majority)
First Write After Heal
Partition failure — time since injection (s)
8s
7s
6s
5s
4s
3s
2s
1s
0s
5.669s
6.573s
6.837s
First Failed Write (Minority Side)
New Leader Elected (Majority Side)
First Write (Majority Side)
Partition heal — time since removal (s)
1.0s
0.9s
0.8s
0.7s
0.6s
0.5s
0.4s
0.3s
0.2s
0.1s
0.0s
0.291s
First Write After Heal

Leader Remains on Majority Side — No Election




Minority Quorum Lost (B-A)
Majority Write Avail. (C-A ≈ 0)
Full Recovery After Heal (F-E)
Minority Quorum Loss (B-A) — per run
Time from partition inject to quorum lost on minority
(1-second polling resolution)
Mean = 4.856s
±1 SD (0.041s)
Quorum Loss Summary
Min: 4.800 s
Mean: 4.856 s
Max: 4.895 s
Std Dev: 0.041 s
Majority disruption: 0 s
(leader stayed on majority)
4.800s
4.873s
4.895s
Time since injection (s)
Run 1
Run 2
Run 3
Partition heal — time since removal (s)
31.0s
31.8s
31.6s
Mean 31.5s
±SD 0.34s
All runs consistent (SD = 0.34 s)
No split-brain — clean ZAB sync
Time since removal (s)
Run 1
Run 2
Run 3

Key Takeaways

Across all three experiments, ZooKeeper behaved consistently with its documented quorum guarantees.

When the leader is killed:

- Election triggered across all runs

When the leader is on the minority side:

- The system self-recovers correctly
- There is a measurable write blackout during leader election

When the leader remains on the majority side:

- The write path is never disrupted
- No election occurs